

The Penrose Class: Personalised Version of the L^AT_EXArticle Class

Documentation and Feature Demonstration for `penrose.cls` Version 2.0

Rashid Talha

Scuola Internazionale Superiore di Studi Avanzati (SISSA), Trieste, Italy

May, 2026

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

1 Introduction

In this example document we describe how to use the `penrose` class. This is not a minimal working example, and might not cover all the features of this class. One should see the source code of this PDF for a more technical introduction. We have also used the `mpt-core` and `mpt-thm` packages written by the author to include brief samples of mathematical content. These are completely optional, and the user can use their own mathematical packages if needed.

2 Metadata and The Title Block

The class provides a flexible way to handle your document information. You can specify a subtitle and affiliation, which are automatically formatted in the title block. Note that the `\title` command accepts an optional argument in square brackets; this is what appears in the page headers from page two onwards, preventing long titles from overflowing the header space.

Similarly, the `\date` command is versatile. If you prefer a formal look, the standard command adds a "Date:" prefix. If you want a cleaner look, you can use the starred version `\date*{\{...\}}` to print the date on its own.

Use `\maketitle` to display the title block. Both the `\maketitle` command and the `abstract` environment are optional and should be used after `\begin{document}`.

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit

in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

3 Tables and Columns

The class extends standard tables by providing custom column types through `tabularx` and `booktabs` packages. These are particularly useful when you need precise control over alignment and width. These are listed in table (1). Additionally the standard column type `l`, `r`, `c`, `p{<length>}`, `X` defined by `tabularx` are also available.

Table 1 – Custom column types	
Type	Description
C	Type X with centre aligned text
L	Type X with left aligned text
R	Type X with right aligned text
D{len}	Type p{len} with centre aligned text
U{len}	Type p{len} with left aligned text
V{len}	Type p{len} with right aligned text

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

4 Section Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

4.1 Subsection Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

Theorem 4.1 (Global Hyperbolicity). *A spacetime (M, g) is globally hyperbolic if it contains a Cauchy surface.*

Equation (1) is known as the Einstein’s field equation of General Relativity.

$$R_{ab} - \frac{1}{2}Rg_{ab} = 8\pi GT_{ab} \tag{1}$$

For citations, the class is pre-configured with `natbib`. And finally here are some example of citations [1], [2], [1, page -1].

$$a^2 + b^2 = c^2 \tag{2a}$$

Lorem ipsum dolor sit amet.

$$a^3 + b^3 = c^3 \tag{2b}$$

Equations (2a) and (2b) are subequations.

4.1.1 Subsubsection Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

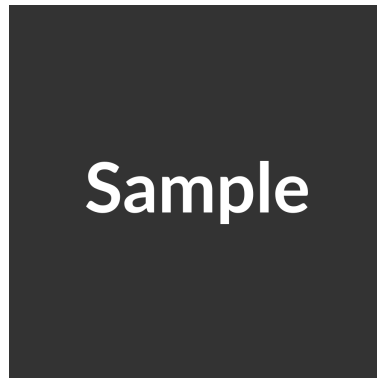


Figure 1 – A sample figure showing the clean captioning style of the class.

4.2 Subsection Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. See figure (1).

5 Section Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

5.1 Subsection Heading

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit

in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim.

In case you want a wider text block. You can use the `widetext` environment.

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

In the other direction, there is a `quote` environment.

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

6 Conclusion

After procrastinating for hours, I finally gave up on avoiding AI and asked ChatGPT to write a concluding paragraph. This is what it had to say: The `penrose` class is designed to be out-of-the-way. It handles the heavy lifting of page geometry and headers so you can focus on the Lie algebras or the gravitational flux. By keeping the design minimal, it ensures that the document remains timeless.

Biblio

- [1] Cohen, P. J. The independence of the continuum hypothesis. *Proceedings of the National Academy of Sciences* **50**, 1143–1148 (1963). <http://dx.doi.org/10.1.1/jpb001>.
- [2] Susskind, L. & Hrabovsky, G. *Classical mechanics: the theoretical minimum* (Penguin Random House, New York, NY, 2014).
- [3] Swetla, M. Canoe tours in Sweden. Distributed at the Stockholm Tourist Office (2015).
- [4] Urry, L. A., Cain, M. L., Wasserman, S. A., Minorsky, P. V. & Reece, J. B. *Photosynthesis*, 187–221 (Pearson, New York, NY, 2016).
- [5] Shapiro, H. M. Flow cytometry: The glass is half full. In Hawley, T. S. & Hawley, R. G. (eds.) *Flow Cytometry Protocols*, 1–10 (Springer, New York, NY, 2018).
- [6] Holleis, P., Wagner, M. & Koolwaaij, J. Studying mobile context-aware social services in the wild. In *Proc. of the 6th Nordic Conf. on Human-Computer Interaction*, NordiCHI, 207–216 (ACM, New York, NY, 2010).
- [7] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria (2018).

- [8] Tang, J. *Spin structure of the nucleon in the asymptotic limit*. Master's thesis, Massachusetts Institute of Technology, Cambridge, MA (1996).
- [9] NASA. Pluto. <https://www.nasa.gov/nh/pluto-the-other-red-planet> (2015). Accessed: 2018-12-06.
- [10] Rempel, R. C. *Relaxation Effects for Coupled Nuclear Spins*. Ph.D. thesis, Stanford University, Stanford, CA (1956).
- [11] Stepney, S. & Verlan, S. (eds.). *Proceedings of the 17th International Conference on Computation and Natural Computation, Fontainebleau, France*, vol. 10867 of *Lecture Notes in Computer Science* (Springer, Cham, Switzerland, 2018).
- [12] Bennett, V., Bowman, K. & Wright, S. Wasatch Solar Project final report. Tech. Rep. DOE-SLC-6903-1, Salt Lake City Corporation, Salt Lake City, UT (2018).
- [13] Suresh, M. Evolution: a revised theory (2006). (in press).